November 12, 1994

Inorganic Cumulative Exam:

There are four questions below, labeled (I) , (II) , (III) , and (IV). ***Answer any one of these. Answer all parts of the question you choose.***

(I) Coordination Chemistry

Define each of the following terms as they apply to coordination chemistry. If possible give an example to illustrate your discussion.

a) Fluxionality
b) Jahn-Teller effect
c) Outer sphere electron transfer
d) Ligand field activation energy in substitution reactions
e) Eighteen electron (or EAN) rule
f) Oxidative addition
g) Pi acceptor (or pi-acid) ligands
h) Linkage isomerism
i) Zero valent complexes
j) Cone angle

(II) Acid-Base Chemistry

a) Briefly describe each of the following acid-base theories. Include a reaction that illustrates the definition.
   1) Arrhenius model
   2) Brønsted-Lowry model
   3) Lewis model

b) Several theories have been developed to explain the relative strengths of acids and bases. Pick one of the approaches and discuss the basis for the theory.
   1) HSAB theory
   2) Drago-Wayland (or E,C) theory
   3) Gutmann (or Donor number) theory
   4) I, B, and F - Strain

c) How are gas phase proton affinities related to pK_b values?

d) Discuss the concept of "solvent leveling" as it pertains to the differentiation of relative acid and base strengths.

e) Occasionally, an acid or base attracts attention because its apparent acidity or basicity falls outside the anticipated range based on homologous molecules. For any one of the following, discuss the origin of its "anomalous" high or low acid or base strength.
   $(H_3Si)_3N$: , H_2CO_3 , H_3PO_3 , HF , 2,6-lutidine

(III) Non-metal Chemistry

In the list below, you will find several important families of non-metallic compounds. Give one or more examples of each and briefly mention any important or unique structural or bonding features of each family.

a) Silicates
b) Aluminosilicates
c) Boranes (neutral)
d) Borohydride anions
e) Carboranes
f) Fullerenes
g) Phosphonitrilic or phosphazene compounds
h) Sulfur-nitrogen rings
i) Silicones
j) Polyhalogens

(IV) Bioinorganic Chemistry

Inorganic complexes are frequently used as models for metallo species found in living systems or as biochemical reagents and drugs. Briefly describe how inorganic compounds have been used in the following areas. Give specific examples as often as possible.

a) O_2 carriers
b) Nitrogen fixation
c) Ion transport
d) Hydrolytic metalloenzymes
e) Iron storage
f) Chelation therapy
g) Chemotherapy
h) Imaging
i) Anti-bacterial agents
j) Electron transfer proteins